Component Name: Commercial airliner wing root attach bracket.
Part Designation: BRK-737-WR-001
Primary Function: Structural load path fitting attaching wing spars to fuselage on a commercial transport aircraft.
Materials Composition: ["7075-T6 aluminum alloy", "anodized finish"]
Mechanical Parameters: Dimensions and tolerances per AS9100 drawing. Limit load capacity 45,000 lbf. Mass 2.1 kg.
Operational Envelope: Temperature range -55C to +70C. Pressurized cabin equivalent to 8,000 ft altitude. Standard comme
Application Interfaces: Bolted to commercial transport fuselage and wing carry-through structure.
Unique Physical Features: Standard civil certified bolt pattern, no stealth features, no combat shock requirements.
Intended Platform: Commercial passenger airliner (civil transport category).
Certification Basis: FAA Part 25 transport category, civil certified.